Tutorial 4

Tutorial Questions

Question 1

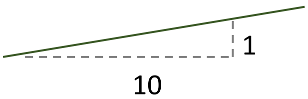
The speed of a train (400 Mg) increases uniformly along a horizontal track from rest to 10 m.s^{-1}, in 100 s. Assuming that the train's engine is responsible for the increase in speed, determine the average power developed.

Question 2

A motor with efficiency of 0.65 is used to lift a 150 kg mass at 1.5 m.s^{-1}. Determine the power input provided to the motor.

Question 3

A 1750 kg car travels up a slope at constant speed. The engine generates a constant power of 37.5 kW, which is transferred to the car with an efficiency of 0.8. Neglecting drag and rolling resistance, determine the car's velocity.



Slope over which the car travels.

Question 4

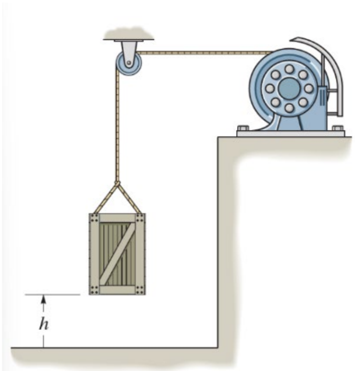
A man, 750 N in weight, runs up a 4.5 m high staircase in 4 s. Determine the power this man generates. Compare the time taken to the time required for a 100 W light bulb to use the same amount of energy.

Question 5

A constant power (1 kW) is supplied to a motor (above). The motor operates with an efficiency, ε = 0.8. Neglecting friction, determine the velocity of a 100 kg crate, with initial velocity 0 m.s^{-1}, after 15 s.

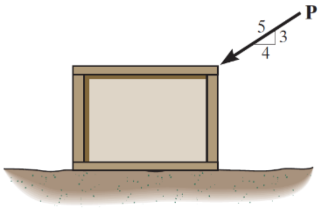
Question 6

A motor hoists a 60-kg crate at a constant velocity to a height of h=5 m in 2 s. if the indicated power of the motor is 3.2 kW, determine the motor's efficiency.



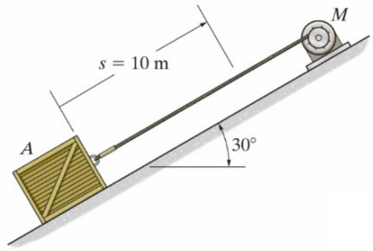
Question 7

If P=200 N, determine the friction developed between the 50-kg crate and the ground. The coefficient of static friction between the crate and the ground is $\mu_s = 0.3$.



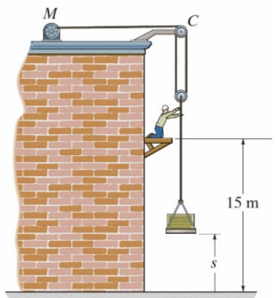
Question 8

If the motor exerts a constant force of 300 N, on the cable, determine the speed of the 20-kg crate when it travels s = 10 m up the plane, starting from rest. The coefficient of kinetic friction between the crate and the plane is μ_k = 0.3.



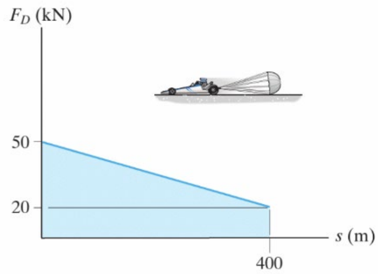
Question 9

If the motor exerts a force of $F = (600 + 2s^2)$ N on the cable, determine the speed of the 100-kg crate when it rises to s = 15 m. The carte is initially at rest on the ground.



Question 10

The 1.8 – Mg dragster is travelling at 125 m/s when the engine is shut off and the parachute is released. If the drag force of the parachute can be approximated by the graph, determine the speed of the dragster when it has travelled 400 m.



Question 11

The driver applies the brakes of a truck, when the light truck travels at 10 km/h. The truck skids 3 m before stopping. How far would the truck skid if it had been travelling at 80 km/h when the brakes were applied?

Question 1

The kinetic energy of the train, after 100 s, is $\frac{1}{2}(400 \times 10^3) \times 10^2 = 2 \times 10^7$ J.

The average power is $\frac{2\times10^7}{100} = 2 \times 10^5$ W or 200 kW.

Question 2

The potential energy increase in 1 s is $150 \times 1.5 \times 9.8$ J.

If the motor were 100% efficient, the power would be $150 \times 1.5 \times 9.8$ W.

Since the efficiency is only 0.65, the power is $\frac{150\times1.5\times9.8}{0.65} = 3.4 \times 10^3$ W or 3.4 kW.

Question 3

The engine does work to increase the potential energy of the car. If the engine were 100% efficient, the car would climb a height of h m in 1 s where

$1750 \times 9.8h = 37.5 \times 1000 \;\Rightarrow\; h = \frac{37.5\times1000}{1750\times9.8}$ m.

However, its efficiency is only 0.8 so the true height climbed is reduced to

$h = \frac{0.8\times37.5\times1000}{1750\times9.8} = 1.75$ m travelled in 1 s, i.e. vertical speed, v_v = 1.75 m.s^{-1}

During this time it travels a horizontal distance of $\frac{10\times0.8\times37.5\times1000}{1750\times9.8}$ m.

So its horizontal speed, v_h = $\frac{10\times0.8\times37.5\times1000}{1750\times9.8} = 17.5$ m.s^{-1}.

Total speed, $v = \sqrt{v_v + v_h} = 17.6$ m.s^{-1}

Question 4

The man generates power to increase his kinetic energy, i.e. the power generated is

$\frac{750\times4.5}{4} = 800$ W or 0.8 kW.

The energy he expends is given by 750 × 4.5 J.

The energy expended by a 100-W light bulb in time t is 100t J. So

$t = \frac{750\times4.5}{100} = 34$ s.

Question 5

The energy supplied to the crate in 15 s is $1000 \times 15 \times 0.8$ J.

If v is the speed of the block, in m.s^{-1}, after 15 s

$\frac{1}{2} \times 100v^2 = 1000 \times 15 \times 0.8 \;\Rightarrow\; v = \sqrt{20 \times 15 \times 0.8} = 15$ m.s^{-1}

Question 6

Power used in lifting crate = $\frac{60\times9.8\times5}{2}$ W.

Power of motor = 3.2×1000 W.

Efficiency = $\frac{60\times9.8\times5}{2\times3.2\times1000} = 0.46$.

Question 7

The magnitude of the normal reaction is given by

$N = (50 \times 10) + \left(\frac{200\times3}{5}\right) = 620$ N,

taking the magnitude of the acceleration due to gravity to be 10 m.s^{-2}.

This means that the maximum frictional force that could be developed would have a magnitude of

$F = 0.3 \times 620 = 186$ N.

But the force tending to move the crate has a magnitude of only $\frac{200\times4}{5} = 160$ N.

This is the magnitude of the force that needs to be developed to prevent the crate from moving; it is less than the maximum possible frictional force, so the crate remains in equilibrium as a result of a frictional force of magnitude 160 N.

Question 8

Force exerted by motor (up the plane) has magnitude = 300 N.

Frictional force (acting down the plane) has magnitude = $(20g \times \cos30°) \times 0.3 = 52$ N.

Weight of crate (acting down plane) has magnitude = $(20g \times \sin30°) = 100$ N.

Magnitude of net force pulling block up slope = 300 – 152 = 148 N.

This is a constant force, giving rise to a constant acceleration of magnitude $a = \frac{148}{20} = 7.4$ m.s^{-2}.

The speed of the block is given by $v^2 = 2as$.

So $v = \sqrt{2 \times 7.4 \times 10} = 12$ m.s^{-1}.

Question 9

Since the cable lift the lower pulley block on both sides, the total force lifting the crate has a magnitude of $2F$.

The work done by this force in lifting the block by 15 m is given by

$W = 2\int_0^{15}(600 + 2s^2)ds = 2 \times \left[600s + \frac{2s^3}{3}\right]_0^{15} = 15 \times 1500$ J.

The potential energy gained by the crate is given by

$U = 100 \times 10 \times 15 = 1000 \times 15$ J.

Thus the kinetic energy imparted to the crate is

$T = W - U = 500 \times 15$ J.

So the speed of the crate is given by $v = \sqrt{\frac{2\times 5\cancel{00}\times 15}{\cancel{100}}} = 12$ m.s^{-1}.

Question 10

The magnitude of the drag force is given, as a function of s, by $D(s) = 50 - \frac{30}{400}s$ kN.

So the work done by the drag force while the car moves 400 m is given by

$W = \int_0^{400} \left(50 - \frac{3}{40}s\right) ds = \left[50s - \frac{3s^2}{80}\right]_0^{400} = 14000$ kJ.

The initial kinetic energy of the car was

$T_i = \frac{1}{2} \times 1.8 \times 125^2 = 14063$ kJ.

So its final kinetic energy is given by

$T_f = T_i - W = 63$ kJ or 63×10^3 J.

Its speed is then given by $v = \sqrt{\frac{2\times 63\times 10^3}{1.8\times 10^3}} = 8.4$ m.s^{-1}.

Question 11

When the car is moving at speed $v_1 = 10$ km/h, its kinetic energy is $T_1 = \frac{1}{2}mv_1^2$.

Assuming that the brakes exert a constant force of magnitude, *F*, the work done by the brakes is 3*F* J, if *F* is measured in newtons.

Thus $3F = T_1 = \frac{1}{2}mv_1^2$.

When the car is moving at speed $v_2 = 80$ km/h, its kinetic energy is $T_2 = \frac{1}{2}mv_2^2$ and the work done by is brakes is $Fs = \left(\frac{1}{3} \times \frac{1}{2}mv_1^2\right)s$.

To stop the car $\left(\frac{1}{3} \times \frac{1}{2}mv_1^2\right)s = T_2 = \frac{1}{2}mv_2^2$.

Thus $s = \frac{3\times v_2^2}{v_1^2} = \frac{3\times 80\times 80}{10\times 10} = 192$ m.